



YILDIZ TEKNİK ÜNİVERSİTESİ MAKİNE FAKÜLTESİ
MEKATRONİK BÖLÜMÜ

INTRODUCTION TO MACHINE LEARNING

UPDATED GUI FOR MACHINE LEARNING METHODS

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Machine Learning Project 1

In this Project we developed a Machine learning Training Interface built using PyQt6. It provides a Graphical User Interface (GUI) where users can experiment with various ML algorithms, load and process data, and visualize results.

General Structure of Code

Libraries Used:

- **PyQt6:** For creating the GUI.
- **scikit-learn:** For classical ML algorithms (Linear Regression, SVM, Decision Tree, etc.).
- **tensorflow/keras:** For deep learning models (MLP, CNN, RNN).
- **matplotlib:** For data visualization.
- **pandas/numpy:** For data processing.

To run code without problems these libraries must be installed. (Compatible Python version: 3.9.2 - 3.9.11)

For tensorflow you must install tensorflow cpu

Main Class: MLCourseGUI

- Inherits from QMainWindow and contains all GUI components.

Data Management

- Libraries that used in code (scikit) provides some libraries but also users can load their own datasets.
- You can adjust ratio between test and train data
- Adjustable parameters for each algorithm (e.g., C, kernel for SVM)
- Data scaling (StandardScaler, MinMaxScaler)

Metrics:

- **Classification:** Accuracy, Confusion Matrix.
- **Regression:** MSE, R^2 Score.

Visualization

Matplotlib Graphs:

- **Regression:** Actual vs. Predicted distribution.

- **Classification:** 2D visualization using PCA

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Example Usage Flow

1. Load Data:

- Select "**Iris Dataset**" from the combo box → `load_dataset()` executes.

2. Choose Algorithm:

- In the "**Random Forest**" tab, set **n_estimators=100**.

3. Train the Model:

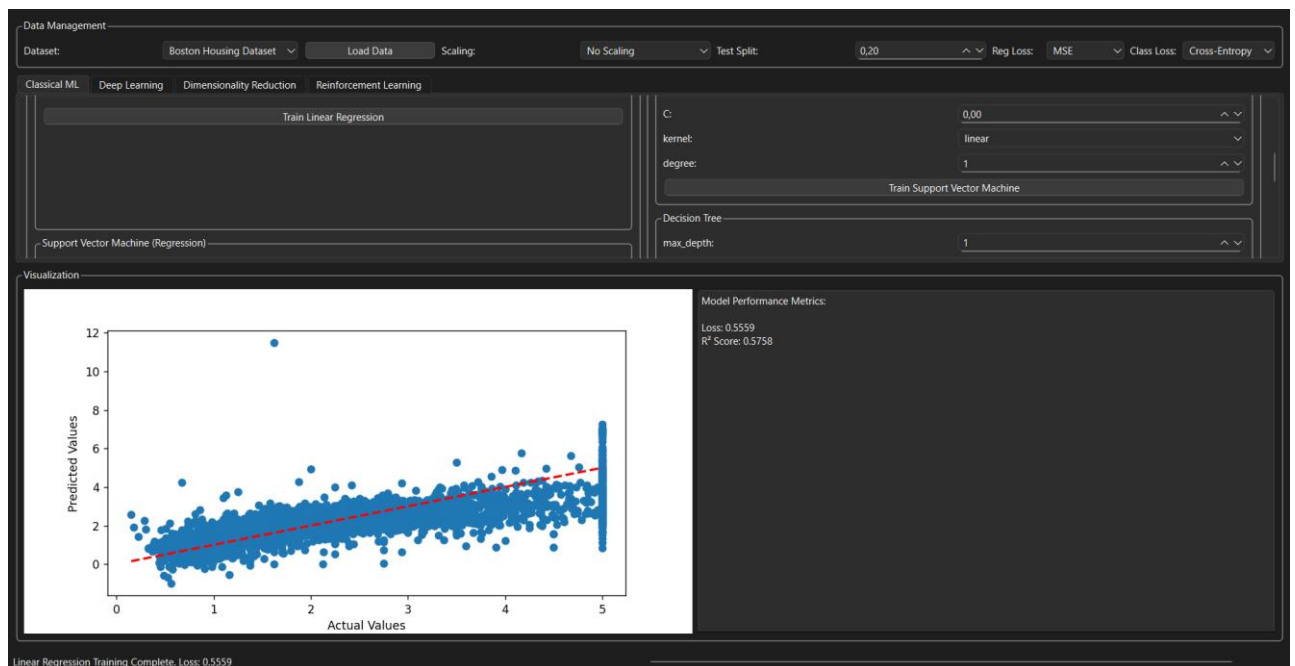
- Click the "**Train**" button → `train_model()` runs.

4. View Results:

- **Accuracy** and **Confusion Matrix** are displayed.
- Graphs are plotted on the canvas.

	In code implement	Dynamic control
Linear Regression	<code>train_model()</code>	<code>loss_regression_combo</code>
Logistic Regression	<code>train_model()</code>	<code>loss_classification_combo</code>
SVM Kernel Seçimi	<code>create_algorithm_group()</code>	kernel ComboBox + degree SpinBox
Naive Bayes	<code>train_model()</code>	<code>var_smoothing SpinBox</code>
Eksik Veri İşleme	haven't implemented	<code>missing_value_combo</code>

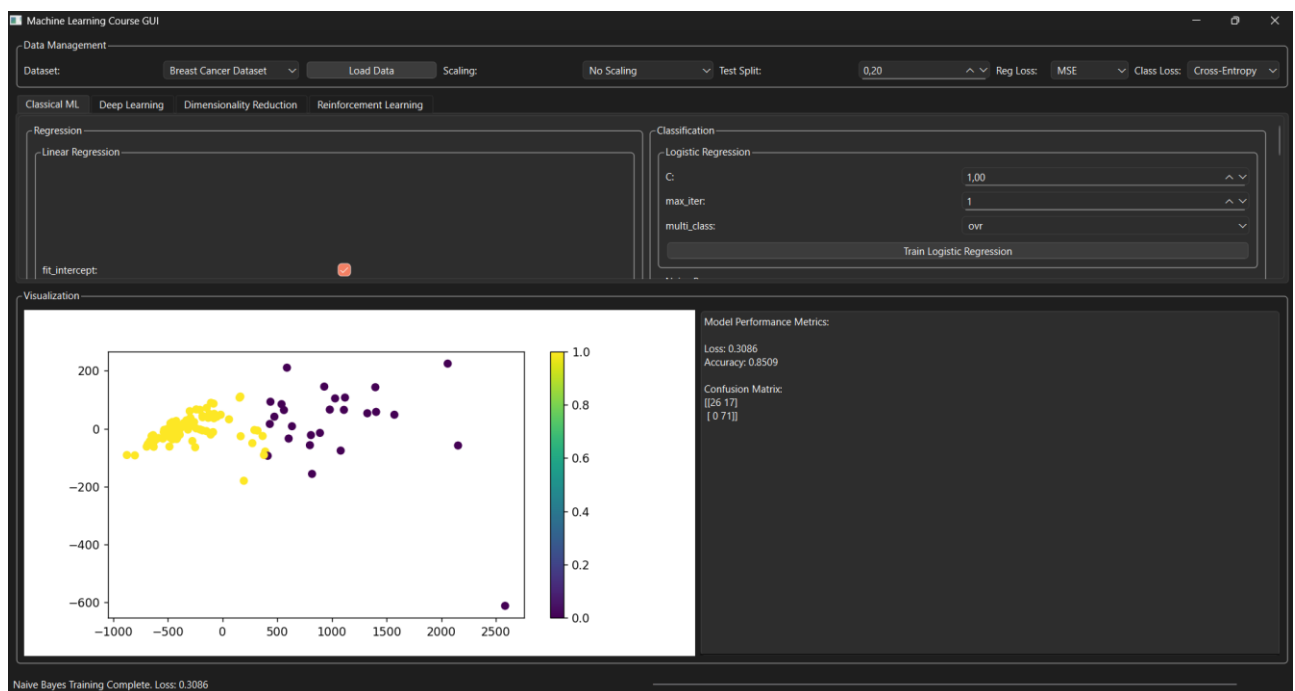
Screenshots from code



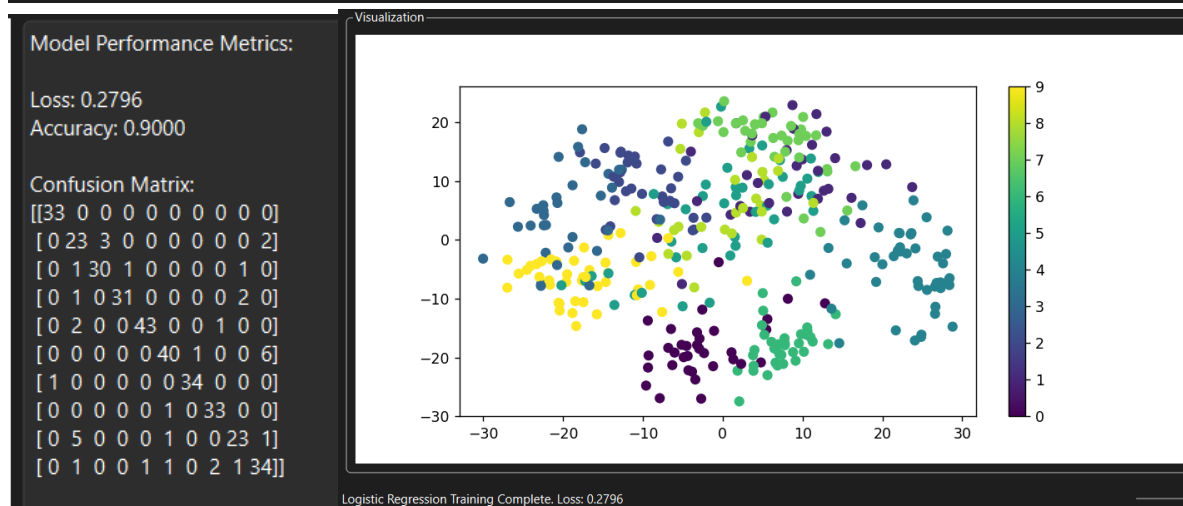
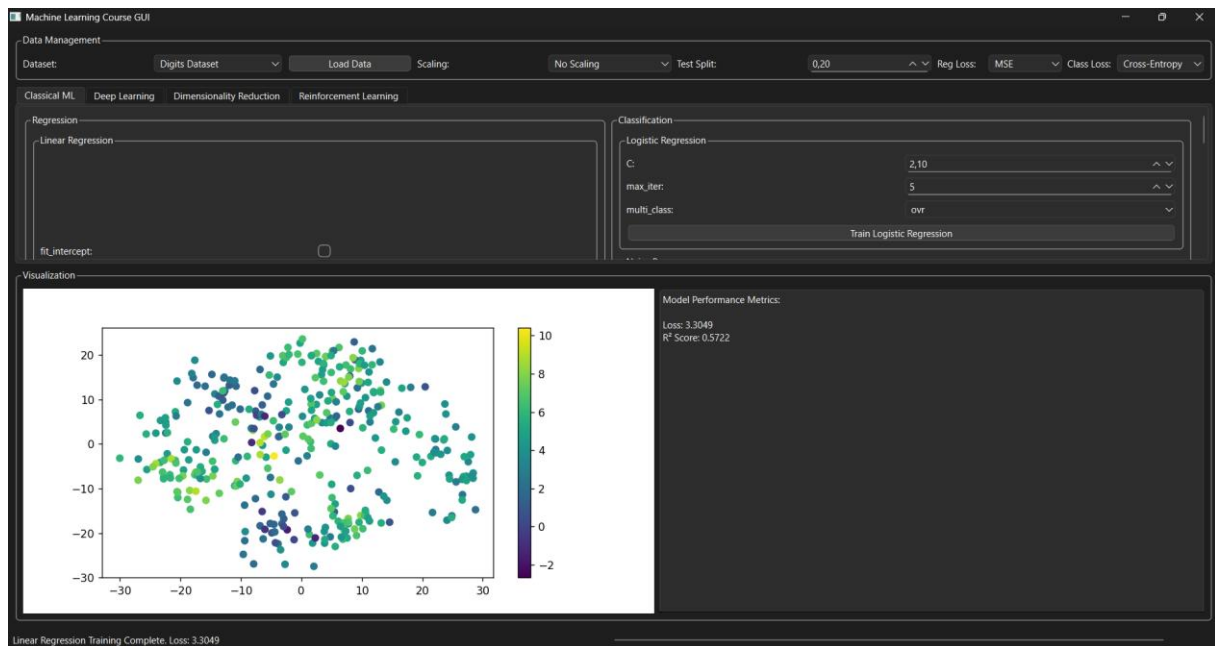
The GUI is almost same, some scrolling problems were fixed. Train models are added.

Boston dataset is updated for California in code.

Here are some screenshots from the program running



Here you can see loss value is between 0 and 1. So loss is acceptable.



Classification

Logistic Regression

C: 1.00 max_iter: 5 multi_class: multinomial Train Logistic Regression